

Cell Design Specification - Power-Tool Battery (case t3_r3)

<i>VBF-T3R3-DS-01. All values are mechanically taken from the parameter set, simulation results, and literature, with per-line sources. Missing items are written "Not provided"; no numbers from memory.</i>

1. Basic Specification

Field	Value	Source
Electrochemical system	NMC811 (LiNi0.8Mn0.1Co0.1O2) / graphite	Chen2020 parameter set (Chen et al., J. Electrochem. Soc. 167 (2020) 080534); deterministic default mapping (task text names no system), recorded in funnel log entry
Nominal capacity	5.0 Ah nominal (parameter set); 4.3176 Ah verified at 1C (DFN simulation)	r5_v12_1c_dfn.json:capacity_ah; criterion >= 2.0 Ah
Voltage window	2.5-4.2 V	pybamm Chen2020 parameter set
Cell dimensions	height 65.0 mm x width 1.580 m unwound strip (wound geometry) x layer stack 153.82 um; shell dimensions Not provided (no shell parameter)	pybamm Chen2020 + params
Electrolyte	EC/EMC + LiPF6 base (c_e0 = 1000 mol/m3); design-lever overrides: kappa = 1.5 S/m, t+ = 0.35, D = 8.0e-10 m2/s; additive candidates: none exercised (no Stage-2 escalation; architecture space judged sufficient - funnel log)	pybamm Chen2020 (base), r5_v12_hip_final_params.json (overrides)
Cation transference number	0.35	r5_v12_hip_final_params.json

2. Electrode and Separator

Layer	Thickness (um)	Porosity	Particle radius (um)	Other	Source
Positive electrode (NMC811)	51.391	0.55	0.80	density 3262 kg/m3	r5_v12_hip_final_params.json / pybamm Chen2020
Negative electrode (graphite)	76.424	0.50	1.00	density 1657 kg/m3	r5_v12_hip_final_params.json / pybamm Chen2020
Separator	12.0	0.50	-	density 397 kg/m3	params override / pybamm Chen2020
Positive current collector (Al)	8	-	-	density 2700 kg/m3	r5_v12_hip_final_params.json
Negative current collector (Cu)	6	-	-	density 8960 kg/m3	r5_v12_hip_final_params.json

N/P ratio: 0.8393 loading ratio (negative/positive areal density) divided by the baseline loading ratio 0.6456 = x1.30 in capacity terms - a +30% negative capacity margin against the Chen2020 baseline stoichiometric parity (mechanical derivation from r5_v12_energy.json:layer_kg_m2 and r1_base_energy.json:layer_kg_m2; per-electrode specific capacities are not separately exposed by the parameter set).

3. Process Design Parameters

Parameter	Formula	Value	Source
Positive areal density	thickness x (1 - porosity) x density	75.44 g/m2	mechanical from params
Negative areal density	thickness x (1 - porosity) x density	63.32 g/m2	mechanical from params
Positive compaction density	density x (1 - porosity)	1.468 g/cm3 (kg/m3 divided by 1000)	mechanical from params

Parameter	Formula	Value	Source
Negative compaction density	density x (1 - porosity)	0.829 g/cm3 (kg/m3 divided by 1000)	mechanical from params
Electrolyte fill amount	pore volume x electrolyte density x fill factor	8.932 g	pore volume from porosity params x 1.2 g/cm3 (literature value, annotated; fill factor 1)
Formation recommendation	e.g. 0.1C CC to 4.2 V, 25 C, 2 cycles	design-recommended value	actual production-line value requires tuning
Thermal management	cooling via h coefficient	25 W/m2/K	r5_v12_hip_final_params.json (design requirement on pack thermal system)

4. Mass Breakdown

Component	Mass (g)	Note
Positive coating (94% NMC811 / 3% CB / 3% PVDF, literature default split)	7.747	layer_kg_m2 x area_m2
Negative coating (94% graphite / 1% CB / 3% CMC+SBR, literature default split)	6.503	layer_kg_m2 x area_m2
Separator	0.245	layer_kg_m2 x area_m2
Positive current collector (Al)	2.218	layer_kg_m2 x area_m2
Negative current collector (Cu)	5.521	layer_kg_m2 x area_m2
Electrolyte (fill)	8.932	pore volume x 1.2 g/cm3 (lit.)
Total stack mass (calc-energy contract caliber, electrolyte excluded)	22.234	r5_v12_energy.json:mass_kg
Total including electrolyte	31.166	mechanical sum

The calc-energy contract formula excludes electrolyte from mass and volume (parameter set lacks electrolyte density); the electrolyte row is provided for completeness.

5. Performance Verification (vs entry-0 criteria)

Metric	Simulated value	Criterion	Determination	Source
Nominal capacity	4.3176 Ah at 1C	>= 2.0 Ah	PASS	r5_v12_1c_dfn.json:capacity_ah
5C discharge retention	98.62% (5C capacity 4.2581 Ah / 1C capacity)	>= 95%	PASS	r5_v12_derived.json:retention_5c
4C fast-charge max temperature	329.75 K (56.60 C)	<= 333.15 K (60 C)	PASS	r5_v12_4c_dfn.json:T_max_K
5C discharge max temperature	319.57 K (46.42 C)	<= 333.15 K	PASS (informational)	r5_v12_5c_dfn.json:T_max_K
Lithium plating at 4C	anode potential min +41.7 mV	< 0 V -> plated	PASS (no plating)	r5_v12_4c_dfn.json:anode_potential_v
Power density	181,557 W/kg	>= 4,000 W/kg	PASS (non-binding)	r5_v12_energy.json:power_density_w_kg
Energy density	706.3 Wh/kg	not thresholded (information)	-	r5_v12_energy.json:energy_density_wh_kg

6. Design Notes

Round-by-round changes with evaluate-log citations (all values DFN judge grade at round 5):

R1 baseline (evaluate_r1_Chen2020-baseline = fail): Chen2020 default thick-electrode energy cell is polarization-limited: 5C retention 8.7%, anode min -537 mV (plating), T_max 354 K, 4C CC acceptance 0.176 Ah.

R2 V1-V3 (evaluate_r2, all fail): porosity +0.05 and loading scale 0.55 -> retention reaches 95.0-98.2% but 4C T_max remains 340+ K; particle refinement 1.5 um alone insufficient.

R3 V4-V7 (evaluate_r3, all fail): N/P x1.3 (V4) gives clean no-plating margin (+36 mV); h=20 cooling (V5) clears 5C T_max (325.3 K); combos V6/V7 push 4C acceptance to 0.774 Ah but 4C T_max 334.2-334.5 K still over.

R4 V8-V11 (evaluate_r4): fine positive particles 0.8 um (V8, T_max 332.65 K - 0.5 K margin), fast D 8e-10 alone insufficient (V9, 333.76 K), h=25 alone passes (V10, 332.34 K); combo V11 champion at screening: T_max 329.87 K, retention 98.5%, anode +39.6 mV, acceptance 0.729 Ah.

R5 judge grade (evaluate_r5, both pass): V11 re-verified at DFN (329.85 K, +37.9 mV); V12-HiP-final combines the V7 high-porosity loading profile with V11 refinements: T_max 329.75 K, anode +41.7 mV, acceptance 0.811 Ah, capacity 4.318 Ah -> champion.

Honest residual: at 4C CC charge the 4.2 V ceiling is reached after 0.811 Ah (effective ~4.6C vs the 5.0 Ah nominal); the no-plating criterion is judged on anode-potential evidence, and charging beyond the window was not simulated.